

Shokunin AI Ecosystem

Enterprise White Paper v4.2.2

github.com/EliasOulkadi/shokunin | 62 Skills | Windows + Linux | Open Source MIT

Executive Summary

Shokunin is a production-grade AI coding ecosystem for Windows 11 and Linux. It provides persistent memory via ChromaDB, 62 specialized AI skills, MCP servers for tool integration, and a declarative self-update system. Zero servers. Zero API costs. Fully offline capable.

Architecture

The system is built around a three-layer memory architecture:

- ChromaDB vector store: semantic search with cosine/euclidean distance for context-aware retrieval across sessions
- JSONL transaction log: structured, parseable record of every interaction for programmatic analysis
- Markdown session files: human-readable summaries for manual review and debugging

The MCP (Model Context Protocol) server implements JSON-RPC 2.0 over stdin/stdout with 8 tools: `store_context`, `search_context`, `get_session_summary`, `multi_search_context` (vector + BM25 + temporal fusion), `consolidate_memories`, `list_sessions`, `continue_session`, and `save_message`.

Skills Ecosystem -- 62 Skills, 10 Domains

Every skill includes YAML frontmatter with name and description for automatic trigger matching. Skills cover: Infrastructure (docker, kubernetes, terraform, ci-cd, db-admin), Backend (api-forge, auth-architect, db-sculptor, error-handler), Frontend (13 design/UI skills), Mobile (flutter, react-native), Quality (test-commander, performance-profiler, code review variants), Content (communication, content-marketing, seo-geo, translate-craft), Productivity (git-workflow, windows-powershell, strategy, finance, legal), Documents (kami, portfolio-auto), AI Agents (agent-browser, agent-tools, skill-creator), Ecosystem (shokunin-update, chromadb, memory).

Production Quality -- v4.2.2

- Python: 100% type-annotated (from `__future__` import annotations), lazy ChromaDB initialization (no module-level side effects), `_dispatch` pattern for JSON-RPC routing, named `_LOGGER` pattern
- PowerShell: `Set-StrictMode -Version Latest` on all 12 scripts, `ErrorActionPreference = Stop`, `CmdletBinding` with comment-based help
- Testing: 29 tests covering security (path traversal sanitization), MCP protocol compliance (tools/list, invalid methods, tool not found), and ChromaDB integration (save/search/recall/session)
- CI/CD: 3 workflows (CI, Memory Tests, Release), mypy strict type checking, bandit security scanning, dependabot, CODEOWNERS, .editorconfig
- Documentation: ARCHITECTURE.md, CONTRIBUTING.md, CHANGELOG.md, MCP-API.md

- Security: Regex-based path traversal prevention, error message sanitization, all except blocks logged, no secrets in code

Getting Started

Windows one-command install:

```
irm https://raw.githubusercontent.com/EliasOulkadi/shokunin/master/install.ps1 | iex
```

Linux:

```
bash <(curl -sSL https://raw.githubusercontent.com/EliasOulkadi/shokunin/master/install.sh)
```

After installation, open a new terminal and run `opencode`. The ecosystem activates automatically with persistent memory across sessions.

Data Flow

```
User -> opencode.exe -> run-opencode.ps1 (wrapper)
  |-> MCP Server (stdin/stdout JSON-RPC 2.0)
  |   |-> ChromaDB (vector embeddings)
  |   |-> sessions/*.jsonl (transaction log)
  |   |-> sessions/*.md (human-readable)
  |-> chroma-helper.py CLI (save/search/recall/consolidate)
  |-> 62 skills (auto-triggered by context)
  |-> shokunin-update.ps1 (declarative self-update)
```

What Is New in v4.2.2

- Python: Type hints on all functions, lazy ChromaDB init, `_dispatch` pattern, similarity formula fix ($1.0 - \text{dist} \rightarrow 1.0 / (1.0 + \text{dist})$), log directory creation fix
- PowerShell: `StrictMode` + `ErrorActionPreference` + `CmdletBinding` on all scripts
- Security: Regex-based path traversal prevention (`_sanitize_id`), error sanitization in MCP responses
- Testing: 29 integration tests (security, MCP protocol, ChromaDB operations)
- CI/CD: mypy type checking, bandit security scanning, dependabot, CODEOWNERS
- Documentation: ARCHITECTURE.md, CONTRIBUTING.md, CHANGELOG.md, MCP-API.md
- Installers: `install.ps1` `sourceDir` fallback fix, `Read-Host` try/catch; `install.sh` `sed` delimiter fix
- Skills: description frontmatter on all 62 skills, reference notes for skills with external dependencies
- 6 stale files purged, `.ruff_cache/` added to `.gitignore`